

Generative AI's Act Two

From Technology Hammer to Real Value

Sequoia Capital

Sonya Huang, Pat Grady and GPT-4 | September 2023

An analysis of generative AI's transition from novelty demos to
end-to-end solutions solving real human problems

Six Decades of Technological Progress Converged to Enable Generative AI

6 Decades

Moore's Law

Exaflops of compute horsepower to process massive data



4 Decades

The Internet

Trillions of tokens of training data, accelerated by COVID



2 Decades

Mobile & Cloud

A supercomputer in every human's palm, ubiquitous distribution

ChatGPT's launch ignited a Cambrian explosion of innovation not seen since the early internet.

GenAI Startups Surpassed \$1B Revenue Faster Than SaaS Ever Did

>\$1B

in startup revenue — months, not years

ChatGPT

Fastest-growing application in history; strong PMF among students & developers

Midjourney

Hundreds of millions in revenue with a team of just 11 people

Character.ai

Average session time of 2 hours per user —
deep engagement

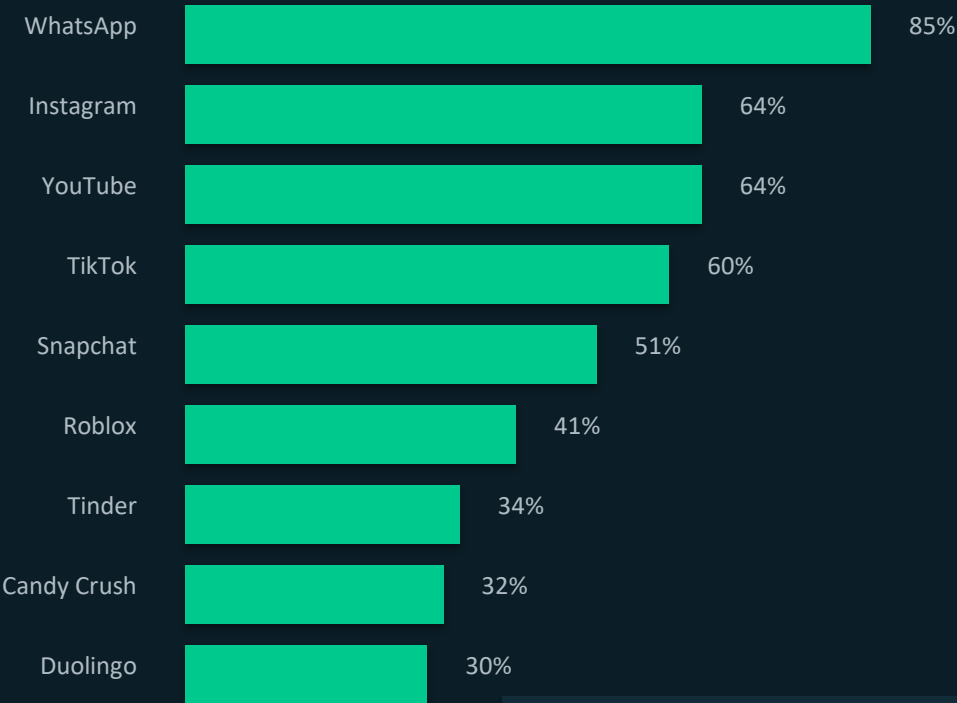
Despite this traction, many AI companies lack product-market fit or sustainable competitive advantage.

AI-First Apps Show 14% Median Retention vs. 51% for Incumbents

Mobile app retention comparison | 3.6x disadvantage for AI-first companies

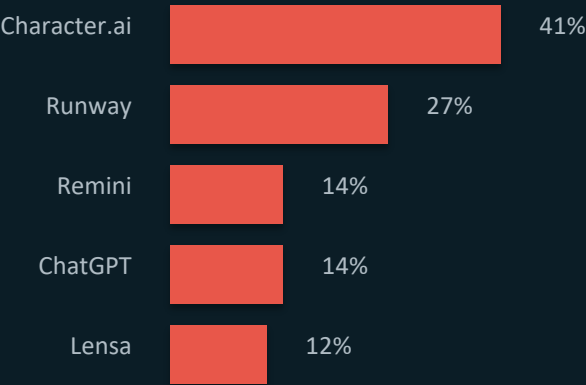
INCUMBENTS

Median: 51%



AI-FIRST

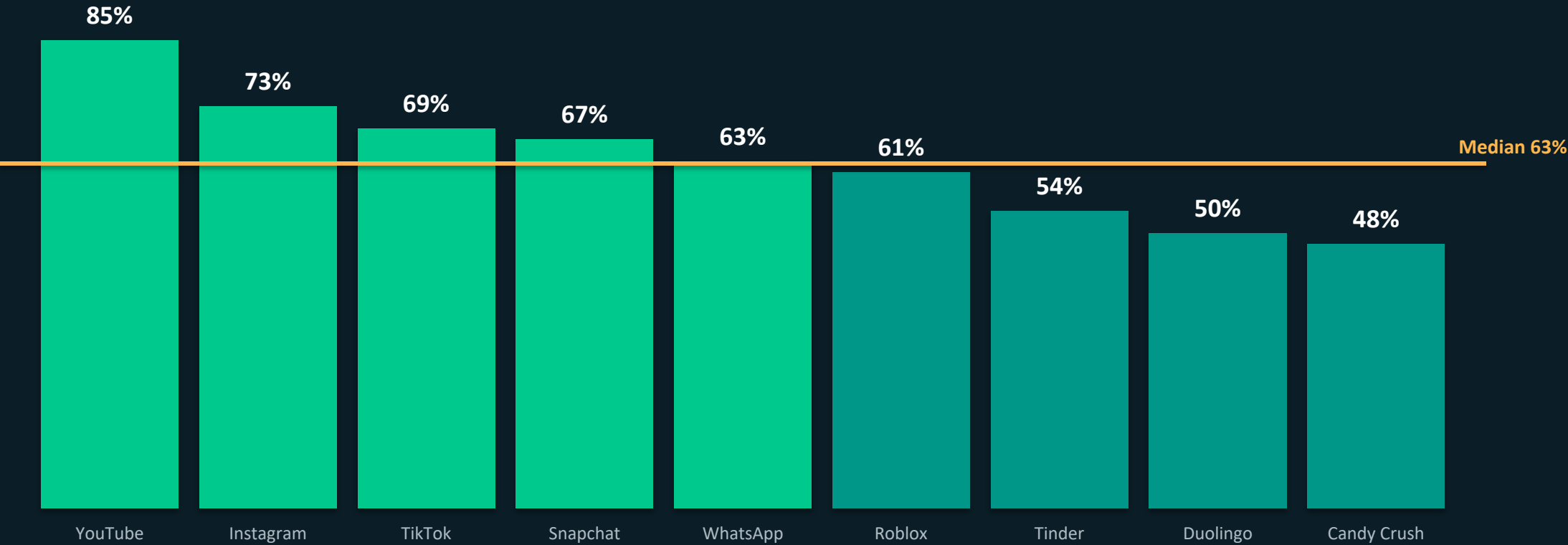
Median: 14%



Character.ai is the retention outlier among AI-first apps at 41%, approaching incumbent levels — suggesting companionship/entertainment may be generative AI's strongest consumer use case.

YouTube Leads Incumbents at 85% Monthly Active Usage

Monthly active usage benchmarks | Median: 63%



Act 1 Was Technology-Out; Act 2 Is Customer-Back

ACT 1: TECHNOLOGY-OUT

"What can this model do?"

- Foundation models as the entire solution
- Novelty demos and lightweight wrappers
- Single-modality applications
- Cool technology in search of a problem
- Poor user retention (14% median)

ACT 2: CUSTOMER-BACK

"What human problem can we solve?"

- Foundation models as a piece of a comprehensive solution
- New editing interfaces for stickier workflows
- Multi-modal in nature
- End-to-end problem solving
- Workflow-integrated, higher retention potential

Act 2 Exemplars: Harvey (legal) · Glean (enterprise search) · Character (companionship) · Ava (digital companions)

Market Map Reorganized by Use Case, Not Model Modality

BEFORE: Organized by Modality

Text Generation

Image Generation

Code Generation

Audio / Speech

Video Generation



NOW: Organized by Use Case

Legal

Harvey

Enterprise Search

Glean

Companionship

Character, Ava

Creative Tools

Multi-modal

Developer Tools

End-to-end

Reflects the evolution from technology hammer to actual value delivery and the increasingly multi-modal nature of applications.

Key Takeaways: What Separates Winners in Act Two

Retention is the critical challenge

AI-first apps at 14% median vs. 51% for incumbents — a 3.6x disadvantage that must be closed.

Foundation models are a component, not the product

Winning companies use models as a piece of a comprehensive, workflow-integrated solution.

Multi-modal, end-to-end solutions will win

New editing interfaces and multi-modal workflows make products stickier and outputs better.

\$1B+ revenue proves real demand exists

GenAI startups hit this milestone faster than SaaS — the market is real despite hype-cycle skepticism.

Solve end-to-end human problems

Companies like Harvey, Glean, Character, and Ava define Act 2 by starting from customer needs, not model capabilities.